PROJECT-BASED LEARNING

PHASE-II EVALUATION

Smart Drone Delivery Management System

Team ID:DSCPP-III-2026-T133

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID:	DSCPP-III-2026-T133
Team Name:	DroneMatrix
Semester:	3rd
Section:	*C / ML-2 / E / J*
Domain:	C++ Drone Delivery

Team Members

1. *Jaiswal, Suryansh – 2028200*
2. *Sahay, Naman – 2028776*
3. *Singh, Nitin – 2027950*
4. *Bhai Patel, Hemang – 2027809*

Mentor

Dr. Hemant Singh Pokhariya **Interactions completed: 3**

Roles: Suryansh – system flow and drone selection; Naman – battery checks; Nitin – request queue; Hemang – testing.

Phase-I Recap & Feedback Incorporated

Phase-I Status

We proposed DroneMatrix, a C++ system that checks whether a drone has enough battery to go from the hub to pickup, deliver the order, and return. Phase-I covered the system design, main data models, and drone-scoring method.

Feedback Received

- *Encouraged creative thinking to improve the project's design and problem-solving approach.*
- *Recommended referring to research papers to identify and justify the formulas used in the project.*

Changes Since Proposal

- *Safety reserve fixed at 20% of mission energy (proposal: 15%)*
- *Unserviceable request is skipped; queue continues*
- *Input validation added for all drone and request fields*

Problem Statement & Finalized Objectives

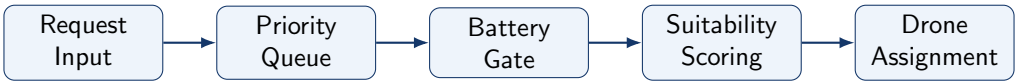
Problem Statement

Road deliveries can be delayed by traffic. Drones can help, but the nearest drone may not have enough battery to deliver and return safely. Our system handles urgent requests first and checks battery reserve before assigning a drone.

Finalized Objectives

1. *Handle high-priority requests first. If priorities match, handle the request that has waited longer.*
2. *Check battery for the full trip, including a safety reserve, before dispatch.*
3. *Choose a suitable drone using distance, energy use, and waiting time.*
4. *Build a city-delivery simulation and compare it with the nearest-drone method in Phase-III.*

Updated System Architecture / Workflow



Major Components

- *Delivery Request Manager (priority_queue)*
- *Drone Manager (unordered_map)*
- *Battery Feasibility Check*
- *Suitability Scoring*

Changes Since Phase-I

- *Reserve set to 20% (was 15% in proposal)*
- *Battery FAIL rejects the drone; request moves on*
- *Charging queue and GPS tracking moved to Phase-III*

Brief Discussion: First, we check each drone's battery. We score only drones that pass, then choose the lowest score. Data is stored in memory, not in a database.

Module Design & Functional Requirements

Major Modules

1. *Delivery Request Manager*
2. *Drone Telemetry Manager*
3. *Battery Feasibility Check*
4. *Suitability Scoring & Assignment*

Functional Requirements

- *Serve highest priority, then longest wait, first*
- *Reject duplicate drone IDs*
- *Require battery $\geq$ mission energy + 20% reserve*
- *Assign the eligible drone with the lowest score*

Module Dependency

The Request Manager sends requests to Assignment. Assignment checks battery and scores eligible drones using data from the Telemetry Manager.

Technology Stack & Development Methodology

Technology Stack

- Programming: *C++*
- Data Structures: *priority_queue*, *unordered_map*, *vector*
- Database / Framework: *None (in-memory)*
- Development Tool: *Visual Studio Code*
- Version Control: *Git / GitHub*

Development Methodology

1. System architecture and data models *(Phase-I)*
2. C++ data structures and scheduling logic *(Phase-II)*
3. Urban delivery simulation *(Phase-III)*
4. Final documentation and test results

Brief Discussion: We use C++ STL containers for request and drone data. City simulation and GPS-style tracking are planned for Phase-III.

Implementation of PBL Course Concepts

Course	Concepts Actually Implemented	Project Module
Data Structures in C	*Heap-based priority_queue (on a vector) for requests; hash table (unordered_map) for drone lookup by ID*	*Request queue; Drone management*
OOPs with C++	*Classes, encapsulated data, constructors, STL containers, custom comparator (functor)*	*All modules*

Discussion: Explain where each listed concept appears in the code. OS and DBMS are not used.

Development & Implementation Progress

Module	Status	Work Completed
Delivery Request Manager	Completed	*Priority queue ordered by priority, then waiting time*
Battery Feasibility Check	Completed	*Round-trip distance, energy, 20% reserve, pass/fail report*
Scoring & Assignment	Completed	*Suitability score, best-drone selection, skip if none fits*
Drone Telemetry Manager	In Progress	*ID, battery, position, availability stored; release, BUSY/CHARGING states pending*
City simulation & benchmark	Pending	*Planned for Phase-III (nearest-first comparison)*

Overall Project Completion: *50%*

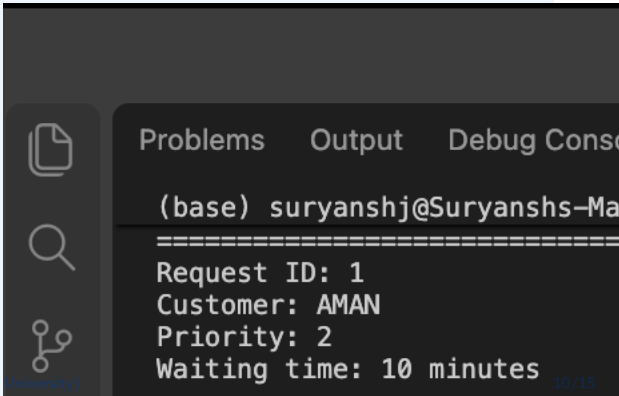
Discussion: Clearly state what is complete, in progress, and still planned.

Working Prototype / Demonstration

Implemented Features

- *Validated input for drones and requests*
- *Priority-based request processing*
- *Battery feasibility report per drone*
- *Best-drone selection with suitability score*

Prototype / Output



Testing, Results & Validation

Test / Module	Expected Result	Observed Result	Status
Battery pass	*Drone 100%, 2%/km at hub; pickup (3,4), drop (6,8): 20 km, 48% needed → PASS*	*The test was reported as successful*	*PASS*
Battery fail	*Same mission, drone at 30% → FAIL, drone rejected*	*The test was reported as successful*	*PASS*
Priority order	*Higher priority request processed first; ties by longer wait*	*The test was reported as successful.*	*PASS*
No suitable drone	*Message shown; next request still processed*	*The request is skipped and processing moves to the next request*	*PASS*

Results / Validation

Three tests were reported as passing. The fourth test requires confirmation because it was initially marked as failed, but after continues test it has also passed.

GitHub Repository & Development Activity

Repository Information

- GitHub Link: *Repository URL*
- README.md: *Updated*
- Contributors: *All Members Added*
- Repository Status: *Updated*
- Branches: *If Applicable*

Development Activity

- *Regular commits maintained*
- *Meaningful commit messages*
- *Individual contribution visible*
- *Latest implementation uploaded*
- *Documentation maintained*

Discussion: Show the README, contributors, and commit history.

Individual Contribution

Member	Role	Technical Contribution	GitHub Commits
Suryansh Jaiswal	*Team Lead*	*System flow, `processAllRequests()`, best-drone selection*	*5*
Naman Sahay	*Member*	*`batteryCheck()`, `calculateDistance()`, 20% reserve*	*5*
Nitin Singh	*Member*	*`RequestComparator`, request queue, drone storage*	*5*
Hemang Bhai Patel	*Member*	*Input validation, test cases, simulation (Phase-III)*	*5*

Challenges, Remaining Work & Phase-III Roadmap

Challenges Faced

- *Sorting requests by priority and waiting time*
- *Continuing to the next request when one cannot be served*
- *Calculating battery use for the full trip with a reserve*

Solutions / Learning

- *A custom comparator for the priority queue*
- *Separate functions for distance, energy, and scoring*

Phase-III Roadmap

1. *Drone release, BUSY/CHARGING states, charging queue*
2. *City-grid delivery simulation*
3. *Benchmark against nearest-first method*
4. *Testing / optimization*
5. *Final documentation*

Expected Outcomes & References

Expected Final Outcomes

- *C++ fleet management engine using STL*
- *Battery-feasibility and suitability-scoring modules*
- *City-grid simulator with nearest-first benchmark*
- *Applicable to e-commerce and courier delivery*
- *Full project documentation and test results*

References

1. *Stroustrup, B., The C++ Programming Language, 4th Ed., 2013*
2. *Dorling, K. et al., "Vehicle Routing Problems for Drone Delivery," IEEE TSMC, 2017*
3. *Cormen, T. H. et al., Introduction to Algorithms, 3rd Ed., 2009*

Thank You
Questions & Suggestions

Discussion: Summarize the expected result and mention the resources used.